

OSI Model

Definition: The Open Systems Interconnection (OSI) model is a conceptual framework introduced by the International Organization for Standardization (ISO) to describe how data is transmitted across a network using a structured seven-layer architecture.

The 7 Layers of the OSI Model

Layer	Name	Primary Role	Typical Examples / Devices
7	Application	Network services closest to the user and applications.	Web browsers, application protocols
6	Presentation	Translation, formatting, encryption/decryption, compression.	JPEG, MPEG, GIF; TLS/SSL commonly associated here in the conceptual model
5	Session	Establishes, maintains, and terminates application sessions.	Session management, checkpoints
4	Transport	End-to-end delivery, segmentation, and reassembly.	TCP, UDP
3	Network	Logical addressing and routing between networks.	IP, routers
2	Data Link	Framing and node-to-node delivery.	MAC addresses, switches, bridges
1	Physical	Transmission of raw bits over physical media.	Cables, hubs, repeaters, modems

7. Application Layer

The Application Layer sits at the top of the OSI stack and serves as the direct interface between software applications and network services. It produces the data that will be sent and handles information received from the lower layers.

- Interacts directly with end-user applications and network-access tools such as Google Chrome.
- Provides network services used by applications to communicate.

6. Presentation Layer

Often called the **Translation Layer**, the Presentation Layer ensures that data is formatted, secured, and compressed so that the receiving application can correctly interpret it.

Key Functions

- **Translation:** Converts application data into a standardized representation for transmission, helping bridge differences in data representation between systems.
- **Standards & Formats:** Handles encoding and representation of media using standards such as JPEG, MPEG, and GIF.
- **Encryption/Decryption:** Describes the conversion of plaintext into ciphertext and back using key values. TLS/SSL is commonly associated with this function in the conceptual OSI mapping.
- **Compression:** Reduces the number of bits required for transmission, improving network efficiency.

5. Session Layer

The Session Layer manages the establishment, maintenance, and termination of connections between applications.

- Helps manage communication sessions between applications.
- Inserts checkpoints into the data stream, allowing a transfer to resume from a saved point after a failure.
- Directs whether communication is **half-duplex** (alternating) or **full-duplex** (simultaneous).

4. Transport Layer

The Transport Layer ensures the end-to-end delivery of entire messages. It acts as a liaison between the Application and Network layers, providing transport services that help data reach the correct application on the destination host.

- **Segmentation & Reassembly:** Splits large messages into smaller segments for transmission and reassembles them at the destination.
- **Protocols:** Common protocols include TCP (reliable) and UDP (fast/lightweight).
- **Connection-Oriented (TCP):** Establishes a connection before data exchange and uses mechanisms such as acknowledgements and retransmission to support reliable delivery.
- **Connectionless (UDP):** Sends datagrams without establishing a formal connection and does not provide a guarantee of delivery or ordering.

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Delivery	Reliability mechanisms	No delivery guarantee
Ordering	Ordered byte stream	No ordering guarantee
Typical focus	Reliability and completeness	Low overhead and speed

3. Network Layer

The Network Layer manages data transmission between hosts across different networks by handling logical addressing and path finding.

- **Data Unit:** Transport-layer segments are encapsulated into packets.
- **Logical Addressing:** Uses IP addresses in packet headers to identify source and destination devices.
- **Routing:** Determines a path for packets to travel across interconnected networks.
- **Hardware:** Primarily implemented through routers.
- **Inter-networking:** Facilitates communication between different networks by directing traffic toward its destination.

2. Data Link Layer

The Data Link Layer serves as the bridge between physical hardware and logical networking, providing node-to-node delivery over a local link. It organizes data into frames and uses link-layer addressing.

- **Data Packaging (Framing):** Data packets received from the Network Layer are organized into manageable units called frames.
- **Addressing and Hardware:** Uses MAC addresses (physical/link-layer addressing) to identify network interfaces and places relevant source and destination MAC addresses in the frame.
- **Devices:** Common devices associated with this layer include switches and bridges.
- **Error Detection:** Link-layer technologies can use mechanisms such as frame checks or CRC to detect transmission errors.

1. Physical Layer

The Physical Layer is the foundation of the OSI model and handles the transmission of raw, unstructured bitstreams over a physical medium from one node to the next.

- **Core Responsibility:** Manages hardware-level transmission of individual bits and converts physical signals into a form that the Data Link Layer can process.
- **Essential Hardware:** Common devices and media include hubs, repeaters, modems, and cables.
- **Bit Synchronization:** Coordinates the timing of bit transmission between sender and receiver.
- **Bit Rate Control:** Defines the transmission rate, measured in bits per second.
- **Physical Topologies:** Includes structural layouts such as Bus, Star, and Mesh.
- **Transmission Mode:** Describes the direction of data flow: Simplex (one-way), Half-Duplex (two-way, one direction at a time), or Full-Duplex (simultaneous two-way communication).

Key Takeaway

The OSI model divides network communication into seven logical layers. Each layer has a specific role, from transmitting raw bits at the Physical Layer to providing network services to applications at the Application Layer.

Quick memory aid: Application → Presentation → Session → Transport → Network → Data Link → Physical
